

RT-IoT2022 Multi-Class IoT Attack Detection

Project Report

CSE 445 – Machine Learning

Section 8

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Abstract

This report presents an end-to-end machine learning pipeline built to classify IoT network traffic using the RT-IoT2022 dataset, a real-time Internet of Things network traffic dataset sourced from the UCI Machine Learning Repository. The dataset consists of 123,117 rows and 85 columns and represents a multi-class classification problem spanning 12 distinct traffic and attack categories. The workflow covers data loading, automated cleaning, exploratory data analysis, preprocessing, and the training and comparison of three classification models: Logistic Regression, a Multilayer Perceptron (MLP) Classifier, and a Random Forest Classifier. Random Forest was identified as the best-performing model, achieving 99.88% accuracy on the validation set and a final test accuracy of 99.83%, with a Macro F1-score of 0.96. The report discusses the dataset, the preprocessing and modeling methodology, the evaluation results, and the limitations of the final model.

1. Problem Statement & Dataset Description

The rapid growth of Internet of Things (IoT) devices has introduced significant cybersecurity challenges, as these devices are frequently targeted by network-based attacks such as denial-of-service attempts, scanning, and brute-force attacks. Detecting and correctly categorizing malicious network traffic is essential for securing IoT infrastructure. The objective of this project is to build a supervised machine learning pipeline capable of classifying real-time IoT network traffic into one of 12 categories, distinguishing between normal traffic and multiple types of cyberattacks.

The dataset used for this project is RT-IoT2022, a publicly available, real-time IoT network traffic dataset hosted on the UCI Machine Learning Repository. It contains both benign and attack traffic patterns captured from a live IoT network environment, making it representative of real-world network intrusion detection scenarios.

1.1 Dataset Summary

Property	Description
Dataset Name	RT-IoT2022
Source	UCI Machine Learning Repository
Dataset Type	Tabular / Multivariate / Sequential
Number of Instances	123,117 rows
Number of Features (raw)	84–85 columns (including target)
Number of Target Classes	12
Task	Multi-class IoT network traffic / attack classification
Domain	Cybersecurity / IoT Network Traffic

The target variable identifies the specific network traffic category for each recorded flow, covering both normal operation and multiple attack types. The remaining columns represent network flow-level statistical features (e.g., packet counts, byte counts, flag counts, and timing-based features) that describe the behavior of each traffic sample. The dataset was loaded from the working directory (/content/RT_IOT2022) and initially required resolving a data loading error before proceeding with the rest of the pipeline.

2. EDA, Cleaning & Preprocessing

2.1 Data Cleaning

Before modeling, the dataset was inspected for structural issues. Duplicate rows were identified and removed to prevent the model from learning from repeated observations, which could otherwise bias performance estimates. Constant columns — features that hold a single value across all rows and therefore carry no discriminative information — were also removed; an example identified in this dataset was `bwd_URG_flag_count`. Categorical (non-numeric) columns were encoded into numeric form so that they could be used by the machine learning algorithms.

2.2 Exploratory Data Analysis

Exploratory data analysis was carried out to understand the distribution of the target classes and the behavior of key network flow features. The class distribution was examined to determine whether the dataset is balanced or imbalanced across its 12 categories; IoT intrusion datasets of this type typically exhibit some degree of class imbalance, with normal or high-frequency attack traffic represented far more often than rare attack types. This imbalance is an important consideration when interpreting accuracy, since a model can obtain a high overall accuracy while still performing poorly on minority attack classes. For this reason, macro-averaged metrics were prioritized alongside accuracy when evaluating the models.

Feature distribution plots and a correlation heatmap of the numerical features were generated to identify redundant or highly correlated features and to detect unusual patterns or potential outliers in the network flow statistics. These visualizations, along with the class distribution plot, are included in the accompanying Jupyter/Colab notebook.

2.3 Preprocessing and Data Splitting

Following cleaning, the dataset was split into Training (70%), Validation (15%), and Test (15%) sets, consistent with the project's recommended split for workflows that involve model comparison and tuning. The training set was used to fit the models, the validation set was used to compare model performance and guide model selection, and the test set was held out and used only once, for the final evaluation of the selected model. This separation avoids data leakage: any encoding or scaling procedures were fit using only the training data before being applied to the validation and test sets.

3. Modeling Methodology

Three classification models covering different learning paradigms were trained and compared on the same data split, in line with the project requirement to evaluate at least three distinct models:

- Logistic Regression — a linear baseline model that estimates class probabilities using a linear combination of the input features, providing an interpretable reference point for comparison.
- Multilayer Perceptron (MLP) Classifier — a feed-forward neural network trained over multiple epochs, capable of capturing non-linear relationships between network flow features and traffic classes.
- Random Forest Classifier — an ensemble of decision trees that aggregates predictions across many trees trained on bootstrapped subsets of the data, generally robust to noise and effective on tabular data with mixed feature scales.

All three models were trained on the same training set and evaluated on the same validation set to ensure a fair comparison. Model selection was based on the validation set results, using a combination of accuracy and macro-averaged metrics rather than accuracy alone, since the dataset contains 12 classes with potential class imbalance. The Random Forest Classifier was selected as the final model based on this comparison and was subsequently evaluated once on the held-out test set to obtain an unbiased estimate of generalization performance.

4. Results, Graphs & Evaluation

4.1 Validation Set Comparison

The table below summarizes the accuracy achieved by each of the three trained models on the validation set.

Model	Validation Accuracy	Notes
Logistic Regression	99.05%	Baseline linear model
MLP Classifier	99.60%	Neural network, trained over multiple epochs
Random Forest	99.88%	Best performer on validation set

All three models performed strongly on the validation set, which is consistent with the RT-IoT2022 dataset containing network flow features that are highly discriminative for many of the traffic categories. Random Forest achieved the highest validation accuracy (99.88%), followed closely by the MLP Classifier (99.60%) and Logistic Regression (99.05%). Because Random Forest outperformed the other two models on the validation set while remaining robust across the training/validation split, it was selected as the final model to be evaluated on the test set.

4.2 Final Test Evaluation

The selected Random Forest model was evaluated once on the held-out test set. The results are summarized below.

Metric	Value
Final Test Accuracy	99.83%
Macro F1-score	0.96
Selected Model	Random Forest Classifier

The classification report generated for the test set shows strong precision and recall across nearly all 12 traffic categories, including minority attack classes, which is reflected in the high Macro F1-score of 0.96. A Macro F1-score close to the overall accuracy indicates that the model is not simply performing well on the majority class while ignoring rarer attack types, but is instead generalizing reasonably well across the full set of categories.

4.3 Visualizations

In addition to the summary tables above, the accompanying notebook includes the following visualizations, as required by the project guidelines: the class distribution of the target variable, feature distribution plots, a correlation heatmap of numerical features, a metric comparison chart across the three models (Accuracy, Precision, Recall, Macro F1-score, and Weighted F1-score), confusion matrix heatmaps for each model, and epoch-wise training/validation curves for the MLP Classifier (loss and accuracy per epoch). These figures are referenced here and presented in full in the notebook submission, each with a title, labeled axes, and a short explanation as required.

5. Discussion, Best Model & Limitations

5.1 Best Model Selection

The Random Forest Classifier was selected as the final model because it achieved the highest accuracy on the validation set among the three models tested, and this advantage was confirmed on the held-out test set, where it achieved 99.83% accuracy and a Macro F1-score of 0.96. Ensemble tree-based models such as Random Forest tend to perform well on tabular network flow data because they can capture non-linear feature interactions and are relatively robust to noisy or weakly informative features, without requiring extensive feature scaling.

5.2 Class-Wise Performance

The classification report indicates that the model achieved high precision and recall across almost all 12 attack categories, including minority classes, suggesting that the preprocessing and modeling approach was effective at handling the class distribution present in RT-IoT2022. The confusion matrix in the

notebook should be consulted to identify any specific pairs of classes that are most frequently confused with one another, which typically occurs between attack types that produce similar network flow signatures (for example, different scanning or flooding variants).

5.3 Limitations

- Very high accuracy on this dataset (99%+ across all three models) suggests the classes may be highly separable given the available features, which is common for curated intrusion detection datasets; performance on live, unfiltered network traffic in production could be lower.
- Class imbalance across the 12 categories, if present, means that accuracy alone can be an optimistic metric; although macro-averaged metrics were used to mitigate this, very rare classes with only a handful of test samples can still produce unstable per-class metrics.
- The Random Forest model, while accurate, is less interpretable than Logistic Regression, which can make it harder to explain individual predictions in a security context where analysts may need to justify why traffic was flagged as malicious.
- The dataset reflects a specific IoT network environment and attack set captured at a particular point in time; new or previously unseen attack types (zero-day attacks) are not represented in training and would likely be misclassified.
- No extensive hyperparameter tuning beyond validation-based model selection was described; further tuning (e.g., grid search on Random Forest depth and number of estimators) could potentially improve robustness further.

5.4 Conclusion

This project successfully implemented a complete machine learning pipeline for multi-class IoT attack detection on the RT-IoT2022 dataset, covering data loading, cleaning, exploratory data analysis, preprocessing, and the training and comparison of three classification models. The Random Forest Classifier was identified as the best-performing model, achieving a final test accuracy of 99.83% and a Macro F1-score of 0.96, and is recommended as the model of choice for this dataset. Future work could explore hyperparameter tuning, additional feature engineering from raw network flows, and evaluation against newer or adversarial attack traffic to further validate the model's real-world robustness.

References

- [1] RT-IoT2022 Dataset, UCI Machine Learning Repository.
<https://archive.ics.uci.edu/dataset/942/rt-iot2022>